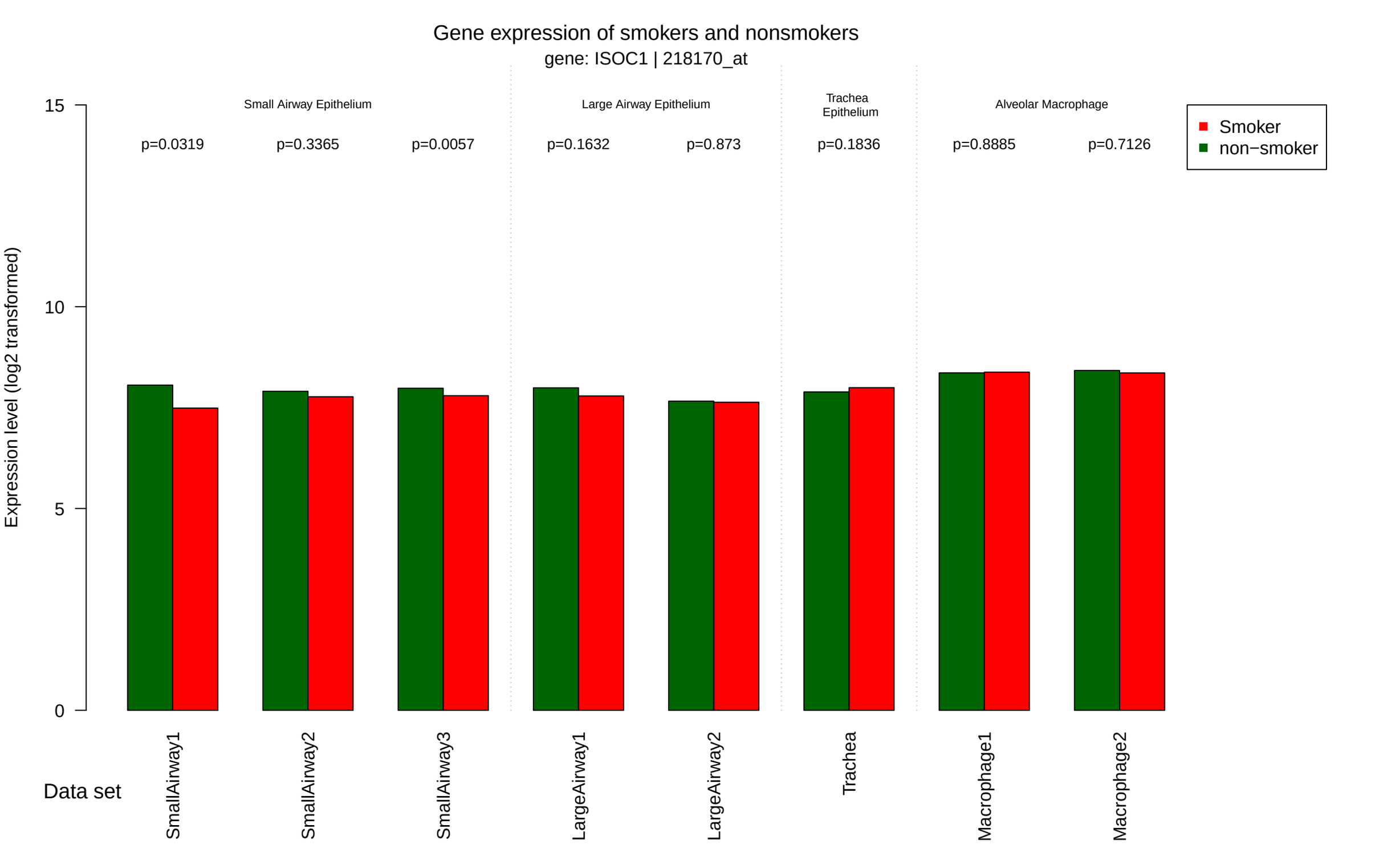
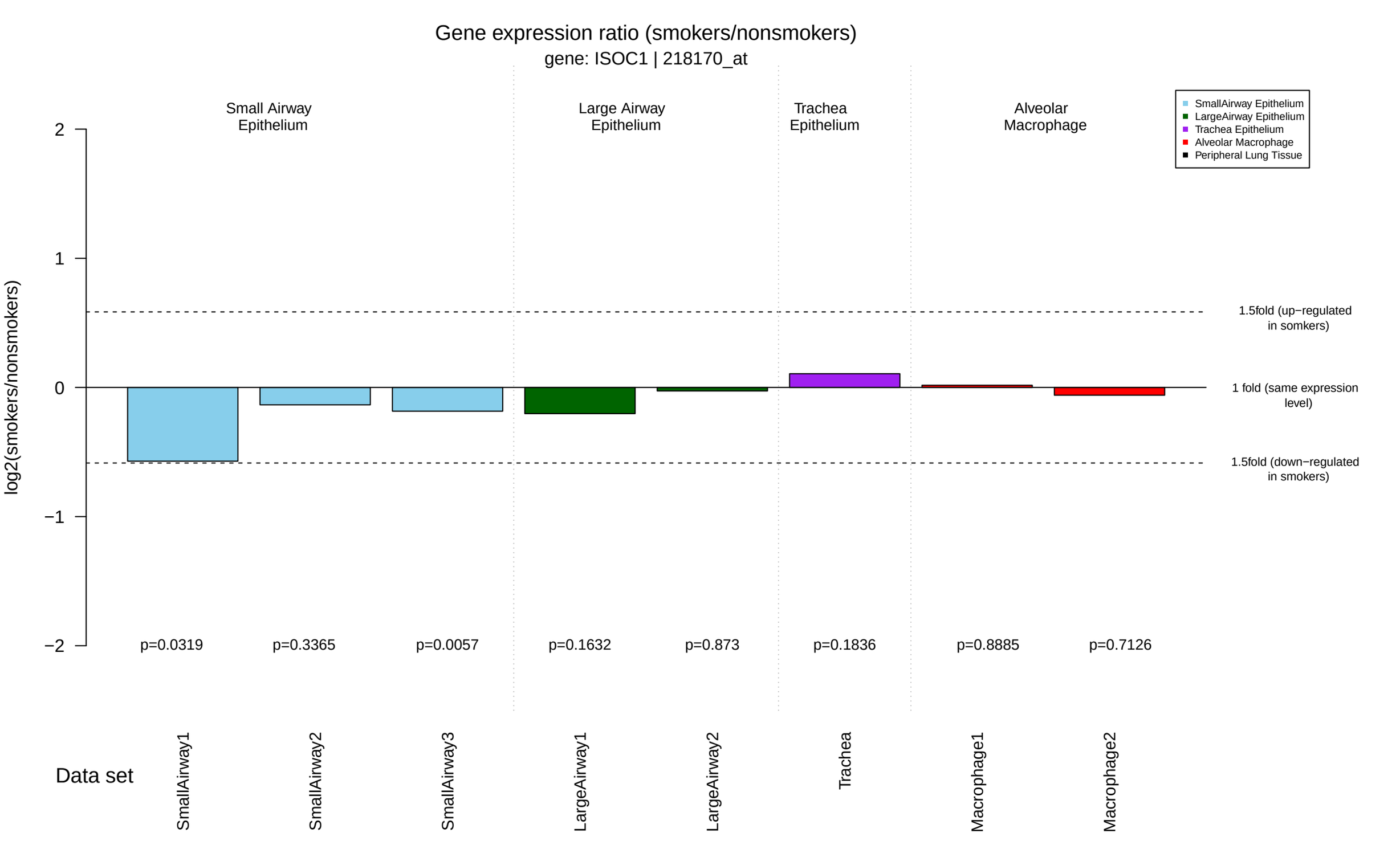
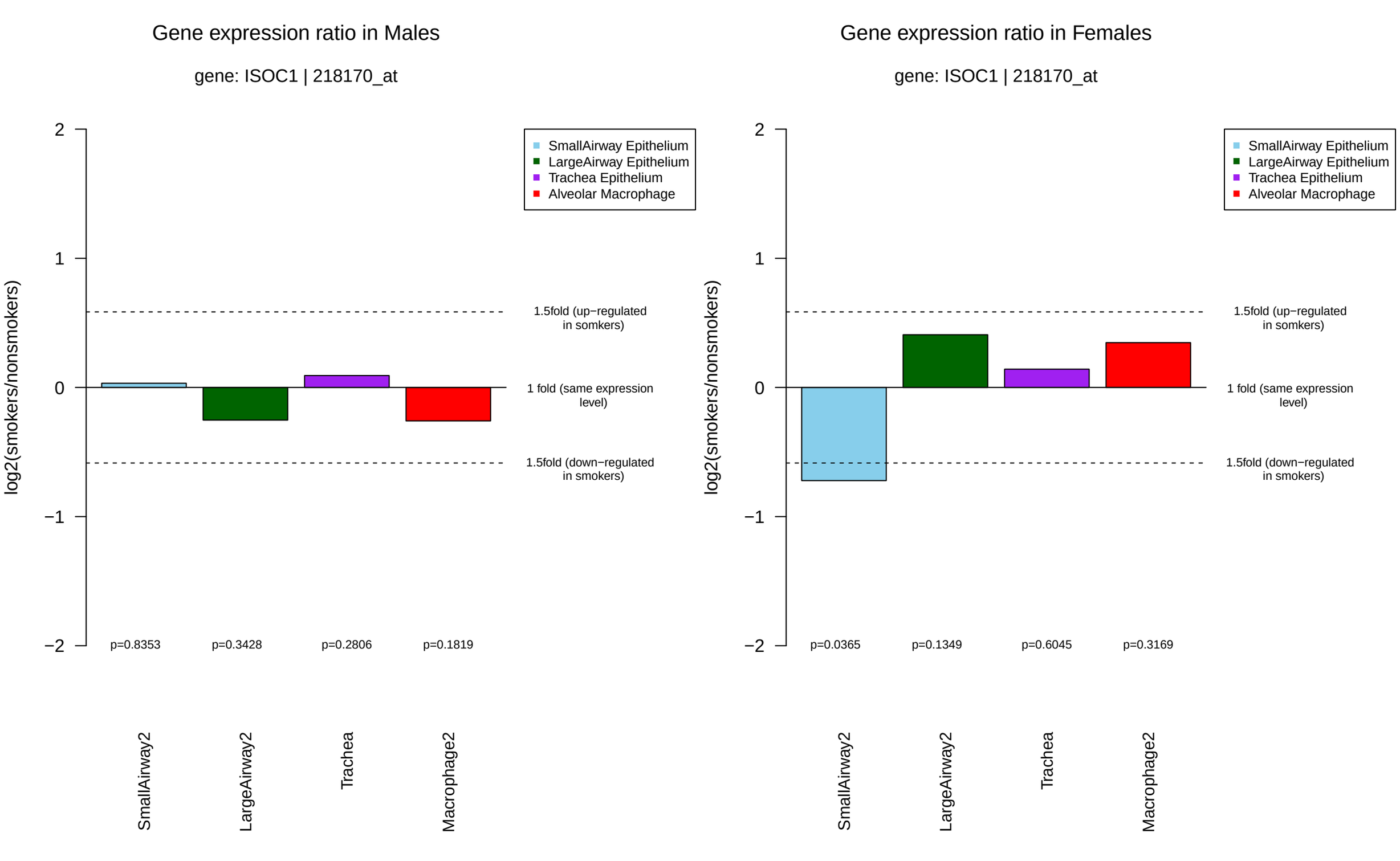
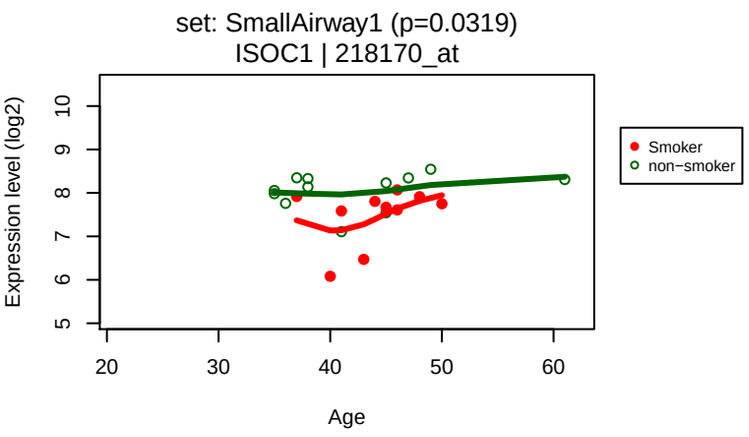
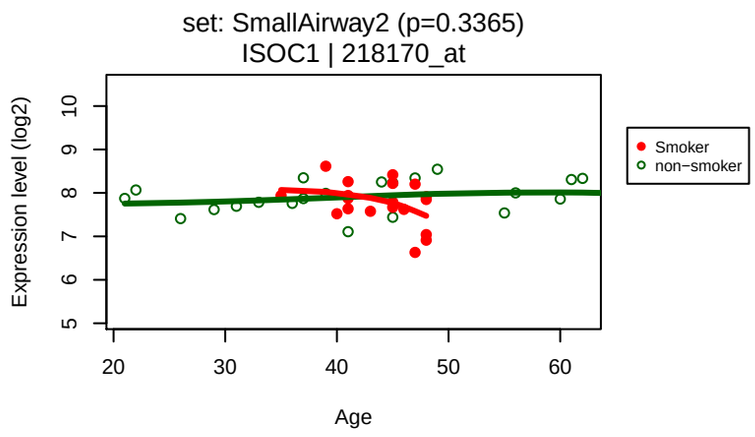
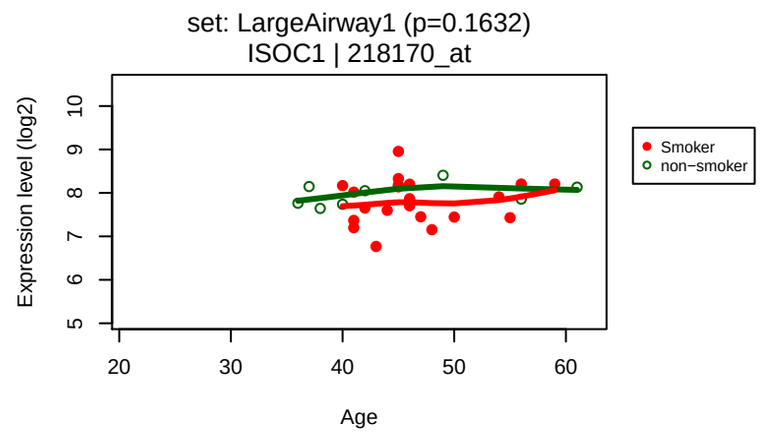

Gene expression of smokers and nonsmokers
gene: ISOC1 | 218170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0319
p=0.3365
p=0.0057
p=0.1632
p=0.873
p=0.1836
p=0.8885
p=0.7126
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

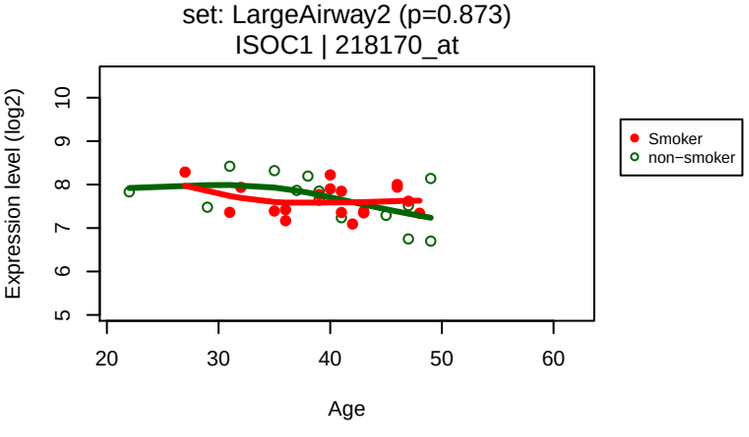

Gene expression ratio (smokers/nonsmokers)
gene: ISOC1 | 218170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0319
p=0.3365
p=0.0057
p=0.1632
p=0.873
p=0.1836
p=0.8885
p=0.7126
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

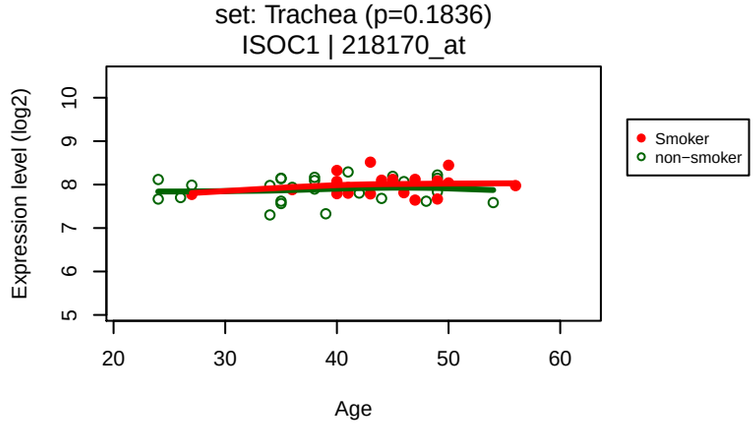

Gene expression ratio in Males
gene: ISOC1 | 218170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8353
p=0.3428
p=0.2806
p=0.1819
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ISOC1 | 218170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0365
p=0.1349
p=0.6045
p=0.3169
SmallAirway2
LargeAirway2
Trachea
Macrophage2

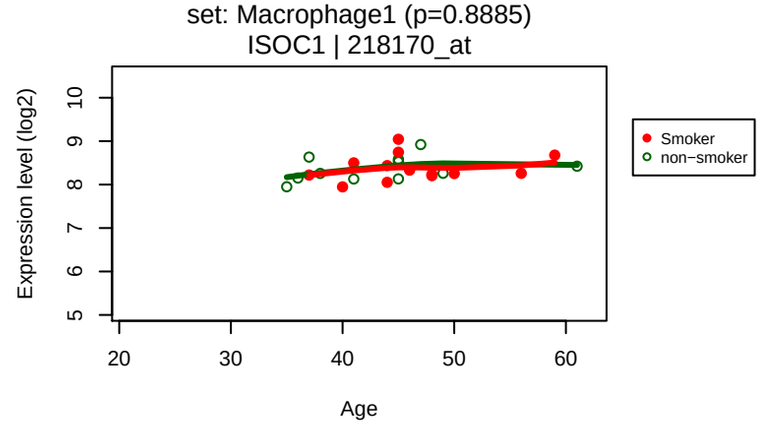

set: SmallAirway1 (p=0.0319)
ISOC1 | 218170_at
Expression level (log2)
Age
Smoker
non−smoker

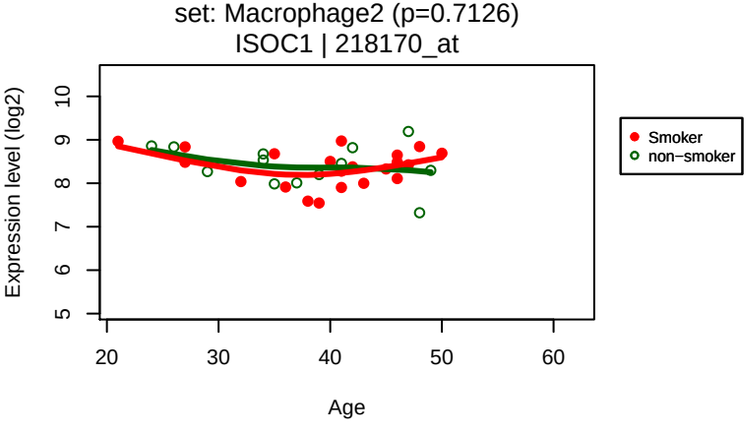

set: SmallAirway2 (p=0.3365)
ISOC1 | 218170_at
Expression level (log2)
Age
Smoker
non−smoker

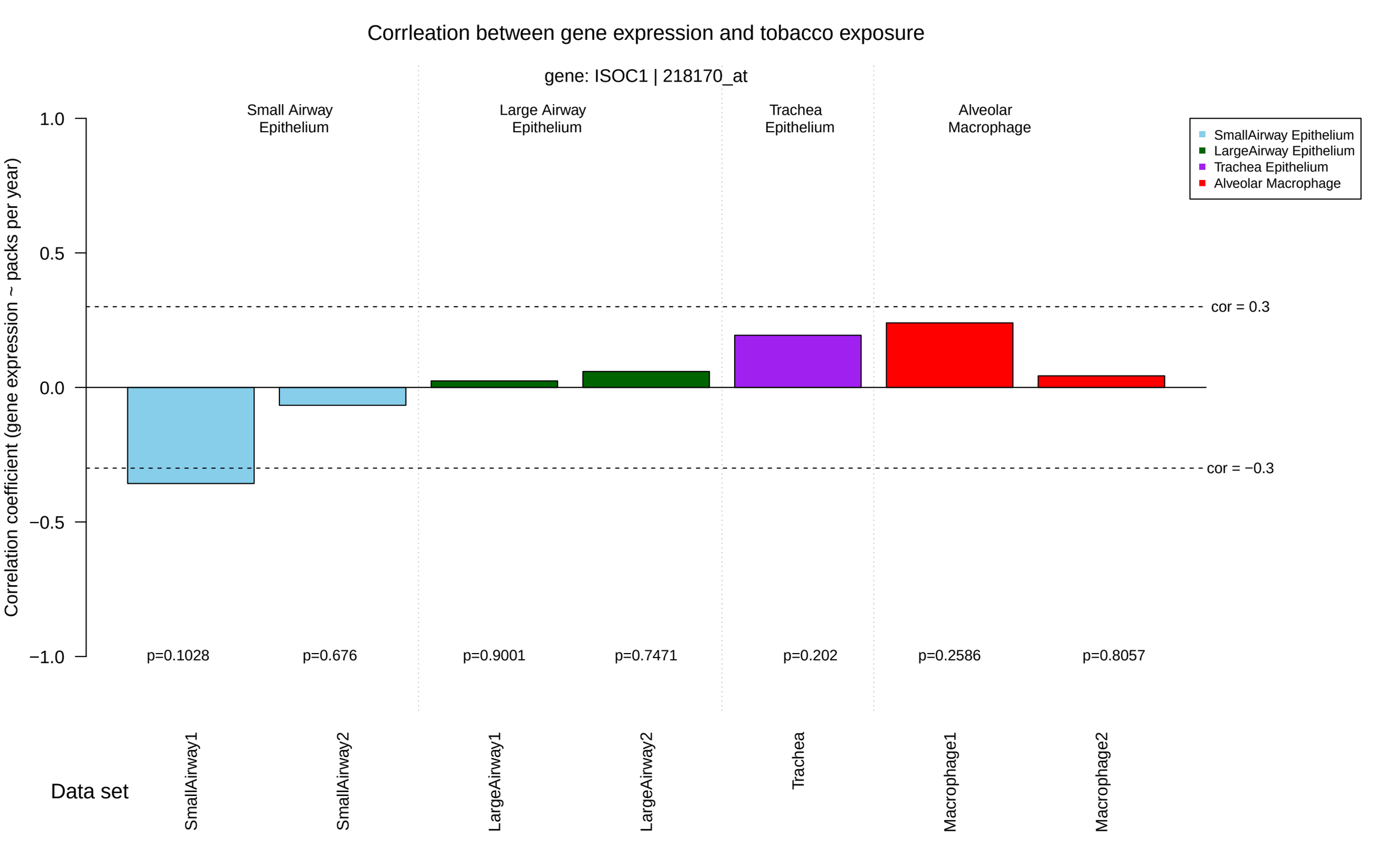

set: LargeAirway1 (p=0.1632)
ISOC1 | 218170_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.873)
ISOC1 | 218170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1836)
ISOC1 | 218170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8885)
ISOC1 | 218170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7126)
ISOC1 | 218170_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ISOC1 | 218170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1028
p=0.676
p=0.9001
p=0.7471
p=0.202
p=0.2586
p=0.8057
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2